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Coordinated control strategy for the short-term frequency response of a DFIG-ES system based on wind speed zone classification and fuzzy logic control

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ABSTRACT

In a power system with a high penetration rate of wind power generation, the rotor speed of a doubly fed induction generator (DFIG) is decoupled from the system frequency; thus, DFIG has no short-term frequency response capability, and the system frequency stability is significantly affected compared to that in a power grid with more conventional synchronous generators. For this reason, a novel fuzzy theory-based coordinated control strategy is presented for improving the short-term frequency response capability of a DFIG-energy storage (ES) system. Firstly, to make the short-term frequency response capability of the DFIG-ES system adaptive for various wind speed conditions, based on the analysis of the DFIG operation characteristics under different wind speeds, four wind speed zones are classified, and accordingly different operation strategies are designed for each wind speed zone. Afterwards, two kinds of fuzzy logic controllers (FLCs) are designed and deployed in the DFIG-ES control system. Specifically, according to the fuzzy logic rules, frequency deviation and DFIG operation state are treated as input parameters, and correspondingly the power support from the combined system and the participation coefficient of the DFIG for short-term frequency support are determined. Moreover, the realization of the presented control strategy is detailedly described. Finally, case studies have been conducted to demonstrate the performance of the proposed control strategy via the comparison between different control strategies, and the simulation results illustrate that the coordinated control strategy can effectively improve the short-term frequency response capability of the DFIG-ES system under various wind speeds.

1. Introduction

With the increasing penetration rate of wind power generation, the intermittence and random features of wind energy have brought great challenges to modern power systems in many aspects [1,2]. Currently, the doubly fed induction generator (DFIG) is the most widely used type of wind turbines (WT) [2]. However, due to the features of electronic converters, the DFIG rotor is indirectly connected to the interconnection point and decoupled from the grid [3,4]. In contrast with the conventional synchronous generators, the rotational kinetic energy of the rotor has been hidden by the electronic converters. As a result, the DFIG does not contribute to grid frequency regulation [5,6] and thus has no short-term frequency support capability to respond to frequency fluctuations. Therefore, in a power grid with large wind power integration, the system inertia will be weakened and the frequency stability will be significantly affected.

In view of the above problems, a series of studies have explored the methodologies of utilizing the kinetic energy of the DFIG rotor for short-term frequency support. Among this body of literature, the addition of another inertial loop to the DFIG control system has been widely used to enhance the inertia of the DFIG [3,4,6–9]. However, this method cannot guarantee a durable and reliable inertial response performance because it is constrained by the operational status of the DFIG and wind power features. Rotor speed control [10–12], pitch angle control [13,14] and the integrated control method [15], have also been studied to improve the short-term frequency support capability of DFIGs. By decreasing the efficiency in the wind energy conversion process and enabling the WT to operate in the de-loading mode, there will be power margin for the DFIG to respond to frequency fluctuations. However, the long-term de-loading operation will obviously reduce the economic benefits of a wind farm. Furthermore, when the DFIG quits the frequency support process, a second decrease in frequency may

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occur due to the sudden power imbalance, resulting in further negative affects to the system frequency stability. Meanwhile, with the development of material science, ES technology has been widely used in mitigating the fluctuations in renewable power generation and enhancing the system stability [16,17]. Flexible controllability, stable power output performance and fast power response are typical characteristics of ES, all of which make it quite suitable for frequency response. In [7], ES is used to increase the inertia of WT. By taking the frequency change rate as an input variable of the ES controller, the ES will take effect in frequency regulation to lift the nadir of the frequency deviations and reduce the frequency drop rate. In the low-inertia grid, when the proportion of WT integration is high, the fluctuations in the WT power output will also cause an apparent mismatch between power generation and consumption, causing a frequency deviation to appear. In [18], the capability of ES for mitigating the power fluctuations of wind farm is explored and ES is used to reduce the frequency deviations by mitigating the WT power fluctuations. However, considering that ES is still expensive, relying solely on ES for frequency regulation will inevitably lead to uneconomical ES sizing results in practice.

Considering the above problems, to exploit the advantages of both the WT and ES in frequency regulation, researchers are beginning to focus on combining the WT and ES in a coordinated way for short-term frequency support. In the previous studies [19–21], the WT maintained a certain amount of frequency reserve capacity through the de-loading operation, and ES was used to compensate for the power shortage when the WT's reserve was insufficient for frequency support. Although this method could meet the frequency regulation requirement for wind farms, the WT still needed long-term de-loading, and it would reduce the economic benefits of the wind farm owners. In [22], a fuzzy-logic based operation strategy for improving the system frequency stability of the WT-ES system was presented. Instead of long-term de-loading, the WT could maintain MPPT mode under this control strategy. With the fuzzy controllers, the ES played a key role in the frequency support. However, the WT and ES in this paper responded to frequency regulation demand separately according to their own operation state. In [23], the fuzzy controllers were designed for the frequency support of the DFIG-ES system, and the frequency deviation Δf and wind speed were set as the input variables of the fuzzy controllers for determining the DFIG output reference. However, the rate of change of frequency (RoCoF) was only used in the comparator block as a reference threshold value and the fuzzy controllers were designed only from the aspect of DFIG power output. Another study in [24] proposed a method that enabled the wind farm to respond to frequency regulation requirements with cooperation between the WT and ES. In the control strategy, the WT could cooperate with the ES dynamically considering the state of charge (SOC) of the ES. Through the response capability assessment, the frequency regulation requirement was properly distributed between the WT and ES, and the WT-ES system had short-term frequency response capability. However, the WT operation state varied with the wind speed; thus, its capability for frequency support also varied under different wind speeds, the control strategy of this literature was only applied and validated under a constant wind speed, and it did not consider scenarios under various wind conditions, which would affect the reliability and performance of the control strategy. Therefore, although the control strategies proposed by the above-mentioned studies did improve the short-term frequency response ability of the DFIG to a certain extent, the rotor kinetic energy of the DFIG in those studies lacks effective coordination with the ES potential on frequency support. Moreover, different operation states of the DFIG under various wind speeds should not be ignored during the process of designing an operation strategy, and the constraints on the economic benefits of a wind farm caused by the long-term de-loading operation of the DFIG in the frequency support process should also be considered.

Considering the above problems and to improve the short-term frequency support capability of a single DFIG, this paper puts forward a fuzzy-based DFIG-ES coordinated control strategy based on the separation of four wind speed zones. The main novelty and contributions of this paper are as follows:

(1) A novel coordinated control strategy of DFIG-ES system for short-term frequency support is designed based on fuzzy logic controllers (FLCs). Under this strategy, the DFIG no longer needs to operate in the long-term de-loading mode for providing frequency reserve, and when there is no frequency support demand, the DFIG can keep operating in the MPPT mode, incurring less negative effect on the wind farm owner's economic benefits.
(2) The DFIG and ES can cooperate dynamically for short-term frequency support to improve the system frequency stability. The operation state of the DFIG and ES have been considered synthetically in the design process of the control strategy, and the operation state of the DFIG and ES can both be directly reflected in the fuzzy controllers. Through the dynamic cooperation between the DFIG and ES, the frequency stability can be improved and the secondary frequency drop problem in the frequency restoration process can be alleviated.
(3) Wind speed zones are classified based on the analysis of DFIG operating features under different wind conditions. Based on the wind speed zone classification, the fuzzy rules and various operation strategies of the DFIG-ES system are designed, and then the DFIG-ES system can provide frequency support in a wider range of wind speed conditions with better reliability and adaptability.

The remainder of the paper is organized as follows. In Section 2, the coordinated control strategy based on a fuzzy logic control method is proposed for the DFIG-ES system. Section 3 shows the simulation results of case studies to verify the effectiveness of the proposed strategy on a simplified power system. Conclusions are given in Section 4.

2. Coordinated operation strategy based on wind speed zone classification and FLCs for short-term frequency support

There are generally three stages in the frequency response process of the power system, namely, the inertial response, primary response and secondary response [4] (i.e., automatic generation control, or AGC). Due to the limited kinetic energy stored in the DFIG rotor and the high investment cost of the ES infrastructure, the DFIG and ES are more appropriate to be applied in the first two stages of frequency response, which is defined as the short-term frequency response process in this paper.

2.1. Classification of wind speed zones

The operational state of the DFIG varies with the wind speed, and its capability for frequency support also varies accordingly at different wind speeds. As a result, the optimal rotor speed and power output of the DFIG under different wind speed scenarios must be analysed and classified during the operation strategy design process to avoid significant impacts on the reliability of the control strategy performance.

The MPPT operation mode is adopted to capture the maximum wind power at the current wind speed. In this mode, the power output of the DFIG is directly related to the current wind speed condition. Moreover, the maximum power output and optimal rotor speed have different values at different wind speeds [13]. The relationship between the operation characteristics of the DFIG and the wind speed is shown in Fig. 1.

The mathematical model of the DFIG MPPT operation is as follows.

$$\begin{cases} P_m = \frac{1}{2}\rho C_p(\lambda, \beta) A v^3 \\ C_p(\lambda, \beta) = 0.22\left(\frac{116}{\lambda_i} - 0.4\beta - 5\right)e^{\frac{-12.5}{\lambda_i}} \\ \lambda = \frac{R\omega_r}{v} \end{cases} \quad (1)$$

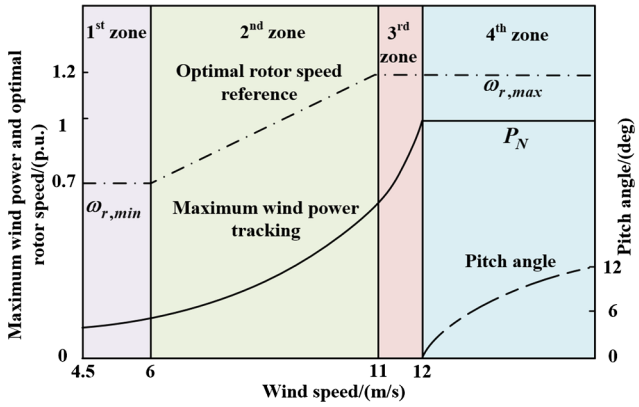


Fig. 1. Operating state diagram of the DFIG wind turbine.

where P_m is the mechanical power extracted from the wind, ρ is air density in kg/m^3, C_p is the power coefficient, λ is the ratio between the rotor blade trip speed and the wind speed, β is the pitch angle, λ_i is the intermediate variable of λ and β, A is area swept by the rotor blades in m^2, v is wind speed in m/s, and ω_r is the rotor speed. When C_p has its maximum value, the DFIG reaches its MPPT point, and the reference power P^*_{mppt} can be expressed as:

$$P^*_{mppt} = k_{opt}\omega_r^3 \tag{2}$$

where k_{opt} is the constant transformed form Eq. (1), when C_p reaches its optimal value, k_{opt} can be calculated. When ω_r reaches its maximum value, the reference power of the DFIG will be set to the rated power output of the DFIG P_N.

According to the analysis of the operation characteristics of the DFIG, four wind speed zones are classified:

(1) The 1st wind speed zone: The interval in which the wind speed is less than 6 m/s is defined as the 1st wind zone. The DFIG rotor speed in this zone can maintain only the minimum rotor speed $\omega_{r,\min}$; thus, it cannot respond to frequency fluctuations with rotational kinetic energy by reducing its rotor speed in this zone.
(2) The 2nd wind speed zone: The wind speed range from 6 m/s to 11 m/s is classified as the 2nd zone, in which the DFIG follows the optimal rotor speed to maintain the MPPT mode for capturing the maximum wind power under the current wind speed. The DFIG can respond to frequency fluctuations with the kinetic energy stored in the rotor except in the scenarios in which the rotor speed is close to its minimum value.
(3) The 3rd wind speed zone: The range of the 3rd wind speed zone is from 11 m/s to 12 m/s. In this zone, the rotor speed will reach its upper limit $\omega_{r,\max}$ and continue to operate at this value; thus, the rotor kinetic energy of DFIG is more sufficient than the 2nd zone. However, since the power output of the DFIG is close to the rated value P_N, the rotational kinetic energy should not be over-released, as doing so may cause the DFIG power limit to be violated.
(4) The 4th wind speed zone: When the wind speed is above 12 m/s, it will be classified as the 4th zone. In this zone, the output power of the DFIG will reach the rated output value P_N, and the control system will maintain the rated output by adjusting the pitch angle. When the DFIG operates in this zone, it must maintain the MPPT mode instead of releasing its rotor kinetic energy. Otherwise, the power limit will be violated.

According to the classification of wind speed zones presented above, the capability of the DFIG for frequency support with rotor kinetic energy at different wind speeds is analysed. However, when the DFIG operates in the 1st and 4th zones, the DFIG does not contribute to the short-term frequency support; therefore, due to the continuously variable nature of the wind speed, the DFIG cannot assure a reliable frequency response capability in its operation, and the wind speed range in which the DFIG can respond to frequency fluctuations is limited. To address this problem, ES is used in this paper to cooperate with the DFIG to expand the frequency support capability over a wider wind speed range based on the wind zone classification. Dynamic cooperation between the ES and DFIG is realized through the FLCs.

It is worth noting that, when the system frequency is above the normal range, through such methodologies as rotor speed control or pitch angle control, the DFIG power output can be decreased by reducing the efficiency of wind energy conversion to make the DFIG operate in a non-optimal mode. The load-reduction requirement for high-frequency incidents then can be fulfilled in most wind zones, and it is easier to reduce the generation than to increase it for supporting low-frequency incidents; thus, this paper mainly focuses on analysing a low-frequency incident.

2.2. FLCs

The fuzzy logic control method does not require a precise mathematical model of the controlled process [23]. When the controlled process is complex or its mathematical model is difficult to establish, the fuzzy logic methodology is a practical method of control. Since the DFIG-ES system power variation model for frequency support is nonlinear and the relations between the controlled variables of the DFIG-ES system and the frequency deviation are complicated, the fuzzy logic control method is chosen as the coordinated controller in this paper.

On basis of the wind speed zone classification presented in Section 2.1, the fuzzy logic control method is adopted in the control block of the DFIG-ES system to improve the short-term frequency support capability of the combined system and to realize flexible cooperation between the DFIG and ES. Two kinds of fuzzy controllers are designed, as shown in Figs. 2 and 3. Among them, FLC1 is designed for determining the total active power from the DFIG-ES system for short-term frequency response. In contrast, FLC2 is designed for deducing the participation coefficient of the DFIG for frequency regulation, after which the power reference of the DFIG for frequency support can be derived. Through the dynamic cooperation between these two kinds of fuzzy controllers, the demand power from the combined DFIG-ES system for short-term frequency support can be determined. Moreover, in the process of frequency regulation, the power output of the ES in the frequency response process can be determined indirectly by FLC1 and FLC2 together. Cooperation of fuzzy controllers can not only make full use of the potential of the DFIG kinetic energy in the frequency response, but also realize the complementary cooperation between the DFIG and ES.

(1) FLC1: FLC1 determines the total short-term frequency power demand of the DFIG-ES system according to the changes in the system frequency. The inertial frequency response features of the synchronous generator can be dynamically simulated in this process. The input variables are the frequency deviation Δf and system frequency change rate df/dt, which have ranges of $[-1, 0]$ Hz and $[-0.6, 0.6]$ Hz/s [22], respectively. The output variable $\Delta P_{total,SPT}$ (0–0.25p.u.) of FLC1 is the total active power increase of the DFIG-ES system for the short-term frequency response, and its range is [0, 0.25] p.u.

The membership functions of the input and output variables of FLC1 are shown in Fig. 2. Seven fuzzy sets, i.e., VS (very small), MS (middle small), S (small), M (middle), L (large), ML (middle large), and VL (very large), are designed to describe the state of Δf and $\Delta P_{total,SPT}$, and another seven fuzzy sets, i.e., NL (negative large), NM (negative middle), NS (negative small), Z (zero), PS (positive small), PM (positive middle), and PL (positive large), are designed to describe the state of the frequency change rate df/dt. The fuzzy logic rules of FLC1 are shown in Table 1, and the basic fuzzy rules are established as follows. When Δf is

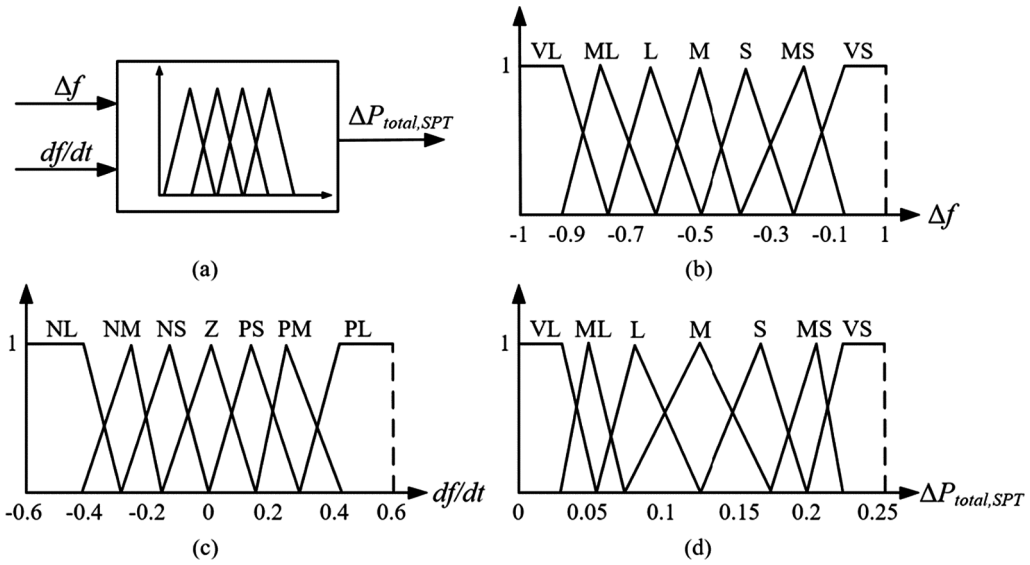


Fig. 2. FLC1 and its membership functions used in the DFIG-ES control system, (a) FLC1 for total active power increment, (b) membership function of Δf, (c) membership function of df/dt, and (d) membership function of $\Delta P_{total,SPT}$.

large, the frequency deviation is still high and the system needs more energy from the DFIG-ES system; thus, the output $\Delta P_{total,SPT}$ will be large. When Δf is moderate and df/dt is large and negative, the frequency is decreasing and the dropping rate should be restrained; thus, the output $\Delta P_{total,SPT}$ will be large. When Δf is small, if df/dt is also small, the frequency does not fluctuate severely or is returning to the normal range from an dramatic fluctuation; thus, $\Delta P_{total,SPT}$ will be moderate or small in the case of excessive regulation. When Δf is large and df/dt is large and positive, the frequency is beginning to return from its nadir to the normal range; thus, $\Delta P_{total,SPT}$ will be large or moderate to help the rising process.

(2) FLC2: For FLC2, the input variables are the DFIG rotor speed ω_r and output power P_w when the DFIG operates in the 2nd and 3rd zones. The ranges of ω_r and P_w are [0.7, 1.2] p.u. and [0.2, 1] p.u., respectively. These two variables reflect the potential of the DFIG for short-term frequency response at the current wind speed. The output of FLC2 is the DFIG short-term frequency response participation coefficient α, which has a range of [0, 1] p.u. The symbol α represents the participation degree of the DFIG for responding to the frequency fluctuations. Therefore, the DFIG power output for frequency support of the combined DFIG-ES system can

Table 1
The fuzzy rule set of FLC1.

df/dt	Δf						
	VL	ML	L	M	S	MS	VS
NL	VL	VL	VL	VL	VL	ML	L
NM	VL	VL	VL	VL	ML	L	L
NS	VL	VL	VL	ML	L	L	M
Z	VL	ML	ML	L	L	M	S
PS	ML	L	L	M	M	S	MS
PM	L	L	M	M	S	S	MS
PL	L	M	M	S	S	MS	VS

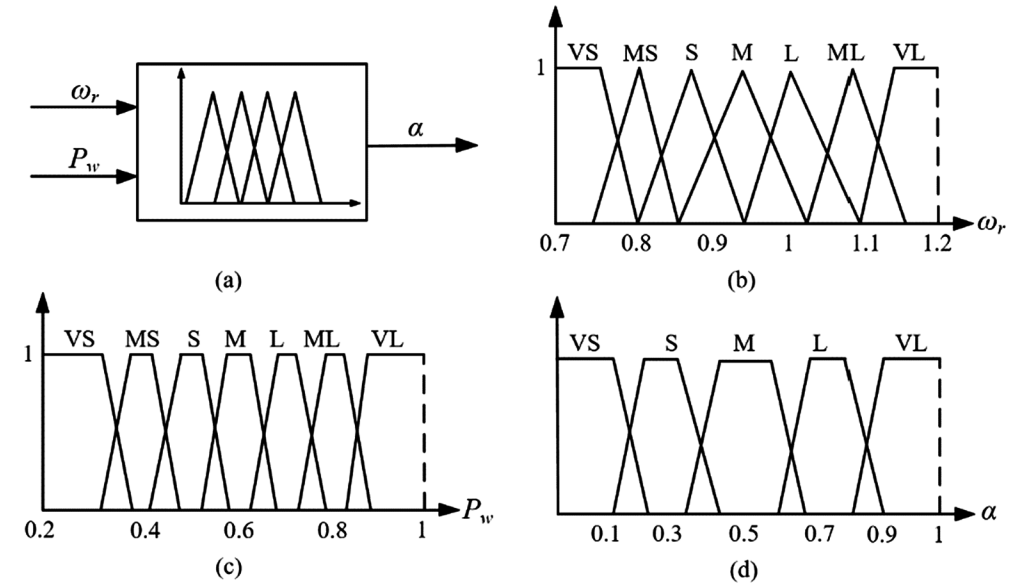


Fig. 3. FLC2 and its membership functions used in the DFIG-ES control system, (a) FLC2 for the DFIG participation coefficient, (b) membership function of ω_r, (c) membership function of P_w, and (d) membership function of α.

be deduced as follows:

$$\Delta P_{w,SPT} = \alpha \cdot \Delta P_{total,SPT} \tag{3}$$

where $\Delta P_{w,SPT}$ is the power variation requirement of the DFIG for frequency regulation. Moreover, the power output for ES can be calculated as follows:

$$P_{ess,SPT} = (1 - \alpha) \cdot \Delta P_{total,SPT} \tag{4}$$

The membership functions of the input and output variables of FLC2 are shown in Fig. 3. The fuzzy sets of FLC2 share the same meaning as for FLC1. According to the wind zone classification in Section 2.1, the basic fuzzy rules for FLC2 are as follows:

When P_w is large, the DFIG power output is close to its upper limit. If the DFIG releases additional energy from the rotor kinetic energy, the power may violate the limitation; thus, the participation coefficient α should be small. When the rotor speed is small, the kinetic energy stored in the rotor is not sufficient for frequency support, and as a result, the output of the FLC2 will be small. When the DFIG power output is small to middle large while the rotor speed is middle to very large, the rotor has sufficient kinetic energy for frequency support and the additional released energy will not exceed the upper power limit; thus, α will be large or very large. The fuzzy rules of FLC2 are shown in Table 2.

By applying two kinds of FLCs in the DFIG-ES system, the DFIG can dynamically determine the participation degree of frequency regulation according to its current operation state through the real-time frequency deviation and change rate, and the DFIG can avoid excessive participation in the frequency regulation process. The total active power output increment of the combined system for frequency regulation is determined by FLC1; thus, the power output for the ES can also be derived from it. The ES can then flexibly cooperate with the operating state of the DFIG, thus realizing the coordinated operation of the DFIG and ES.

2.3. Realization of the DFIG-ES operation control strategy

Based on the classification of wind speed zones and the deployment of the fuzzy controllers in the combined DFIG-ES system, detailed realization of the DFIG-ES operation control strategy is described as follows:

A system frequency drop incident is taken as an example. In the DFIG-ES system, the control module diagram of the DFIG part is shown in Fig. 4. The variables are as follows: T_G is the time constant of the DFIG rotor-side converter, P_e and P_m are the electrical power and mechanical power of the DFIG, respectively, β is the pitch angle, H_w is the inertial constant of the DFIG, T_{cmd} is the torque control signal and P_{cmd} is the power output signal in the torque control mode.

The control process mainly includes three stages: system frequency detection, frequency support and rotor speed recovery.

(1) Frequency detection stage: When the frequency deviation $|\Delta f|$ exceeds the allowable range [24], the frequency regulation flag of the control module will turn to 1, and the combined DFIG-ES system enters the short-term frequency response mode.

(2) Active power-support stage: After entering this stage, the DFIG will exit the MPPT mode, the rotor speed reference value $\omega_{r,ref}$ will be set to track the current rotor speed ω_r (t). According to Section 2.2, by taking the frequency deviation Δf and system frequency change rate df/dt as input variables, FLC1 will output the total active power increase $\Delta P_{total,SPT}$ of the DFIG-ES system; in the meantime, FLC2 will output the participation degree of the DFIG with the DFIG rotor speed ω_r and output power P_w as its input variables, then the DFIG power increment for frequency support $\Delta P_{w,SPT}$ of the combined DFIG-ES system can be determined. After adding $\Delta P_{w,SPT}$ upon the power output signal P_{cmd}, then the increased power output reference of DFIG for frequency support can be obtained. Because the electrical power reference is larger than the optimal mechanical power at this moment, then the rotor speed will decrease and deviate from the optimal value, and the kinetic energy will be released. If the frequency regulation demand cannot be satisfied, then the ES will be used for compensating for the unbalanced regulation demand. With the additional power from the DFIG and the cooperation with the ES, the short-term frequency support capability of the DFIG can be improved.

(3) Speed-recovery stage: When the DFIG active power-support stage is completed, and the frequency regulation demand is fulfilled, the frequency regulation flag will turn to 0, and the DFIG will return to operate at the MPPT mode. At this stage, the optimal rotor speed under the current wind condition will be set as the DFIG reference rotor speed again. The rotor speed will be gradually restored to the optimal speed before the frequency support through the DFIG torque control block. However, in this stage, when the DFIG stops responding to frequency fluctuations, there will be a sudden decrease in the DFIG power output and the second frequency drop incident will occur [21–23]. At this moment, the ES will quickly increase its output to compensate for the power imbalance and reduce the second frequency drop caused by the sudden power decrease in the DFIG output.

Table 2
The fuzzy rule set of FLC2.

P_w	ω_r						
	VS	MS	S	M	L	ML	VL
VS	VS	VS	M	–	–	–	–
MS	VS	VS	L	VL	–	–	–
S	VS	VS	M	VL	VL	VL	–
M	VS	VS	M	L	VL	VL	VL
L	VS	VS	S	M	VL	VL	VL
ML	VS	VS	S	M	L	VL	VL
VL	VS	VS	VS	S	S	VS	VS

As for the ES, one of the main contributions of the ES in the combined system is to cooperate with the DFIG for the purpose of short-term frequency support. Another important function is to help with the DFIG rotor speed restoration and thus avoid the second frequency drop incident. The battery ES model used in this paper is from MATLAB/Simulink original library, and this model is an integrated model of serial and parallel combinations of multiple single cells. The general characteristics of energy storage facilities can be shown in this model. The equivalent circuit of ES is shown in Fig. 5, in which E_{batt} is the nonlinear function of several variables. *Sel* stands for battery operation mode. 0 is the discharge mode and 1 is the charge mode. A is the voltage value (V); B is the battery capacity $(\mathrm{AH})^{-1}$. When the battery is set up as a lead-acid battery, the mathematical model of charging and discharging of the battery is as follows:

Charging mode ($i^* < 0$):

$$f_1(it, i^*, i, Exp) = E_0 - K \cdot \frac{Q}{it + 0.1 \cdot Q} \cdot i^* - K \cdot \frac{Q}{Q - it} \cdot it + Laplace^{-1}\left(\frac{Exp(s)}{Sel(s)} \cdot \frac{1}{s}\right) \tag{5}$$

Discharging mode ($i^* > 0$):

$$f_2(it, i^*, i, Exp) = E_0 - K \cdot \frac{Q}{Q - it} \cdot i^* - K \cdot \frac{Q}{Q - it} \cdot it + Laplace^{-1}\left(\frac{Exp(s)}{Sel(s)} \cdot 0\right) \tag{6}$$

where E_0 is voltage constant (V); K is polarization resistance; i^* is low

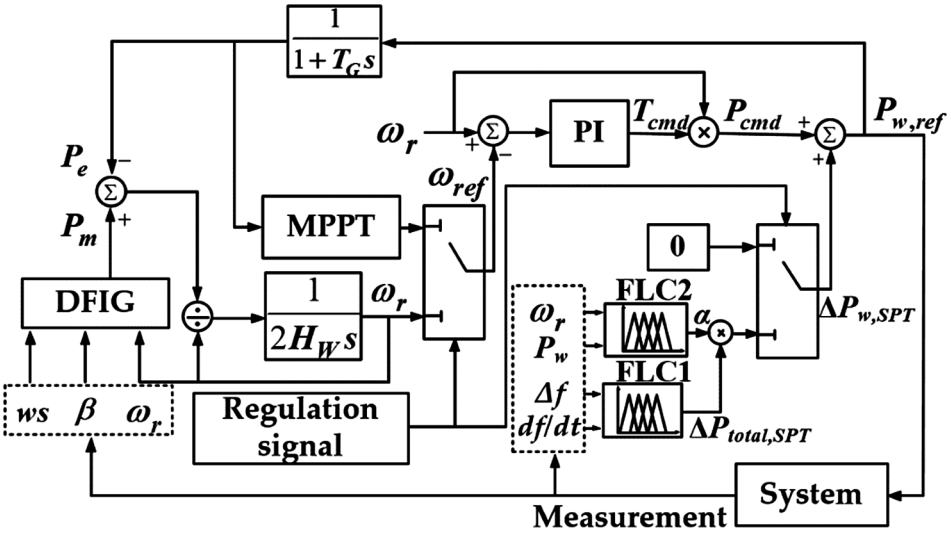


Fig. 4. DFIG control module diagram.

frequency current (A); i is battery current value (A); it is current battery capacity (Ah); Q is the upper limit of battery capacity (Ah); $Exp(s)$ is exponential function; $Sel(s)$ is battery operation mode flag.

The ES control diagram is shown in Fig. 6. P_{ess} is the power output of the ES, $\Delta P_{ess,SPT}$ is the power increment of the ES for frequency support, f_{ref} is the frequency reference value and f is the present system frequency. The ES model used in this paper is the battery module in the MATLAB/Simulink original library.

As shown in Fig. 6, the active power output of the ES for frequency support can be indirectly determined based on the output of FLC1 and FLC2, according to Section 2.2 and (4) above. However, when the frequency is increasing and has nearly returned to the normal range, if the ES output maintains a constant value, the frequency may exceed the normal range instead of staying within it. To avoid an excessive power release of the ES and avoiding the second frequency fluctuation during the frequency recovery process, a droop control module is deployed in the ES control system. When the system frequency returns back above 49.8 Hz [23], the ES output power will gradually decrease. The ES power output reference value is expressed as

$$P_{ess,drop} = -\frac{P_{ess,49.8}(f-49.8)}{f_{ref}-49.8} + P_{ess,49.8} \tag{7}$$

where $P_{ess,49.8}$ is the ES power output when the system frequency reaches 49.8 Hz. To compensate for the impact made by the ES when it stops responding to the frequency fluctuations and ultimately to stabilize the frequency at the rated value, a PI module is added to the control block of the ES unit. This module is enabled when the measured frequency of the system is larger than 49.8 Hz. The ES power output reference value is expressed as

$$P_{ess,ref} = \begin{cases} P_{total,SPT} - P_{w,SPT}, & f < 49.8 \text{ Hz} \\ P_{ess,drop} + \Delta P_{PI}, & f \geqslant 49.8 \text{ Hz} \end{cases} \tag{8}$$

where ΔP_{PI} is the output of the PI controller in Fig. 6, that is, the correction of the ES output power.

Dynamic coordination of the operation of the ES and DFIG is realized through the ES control module and the joint action of the two kinds of fuzzy controllers. Therefore, the DFIG-ES system can support frequency incidents under a wider range of wind speed scenarios. In addition, the second frequency drop caused by the sudden decline of the DFIG output when it exits the frequency regulation can be alleviated.

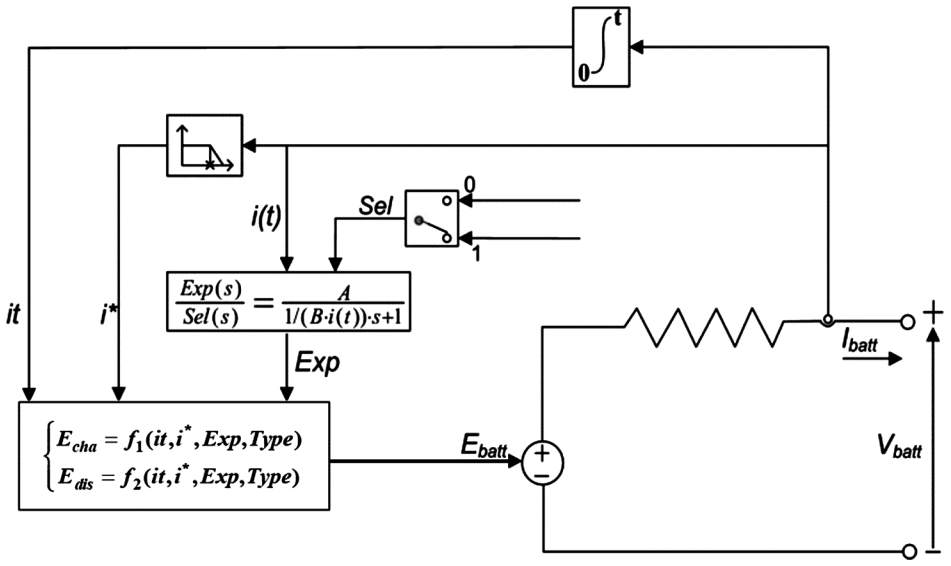


Fig. 5. Battery ES model.

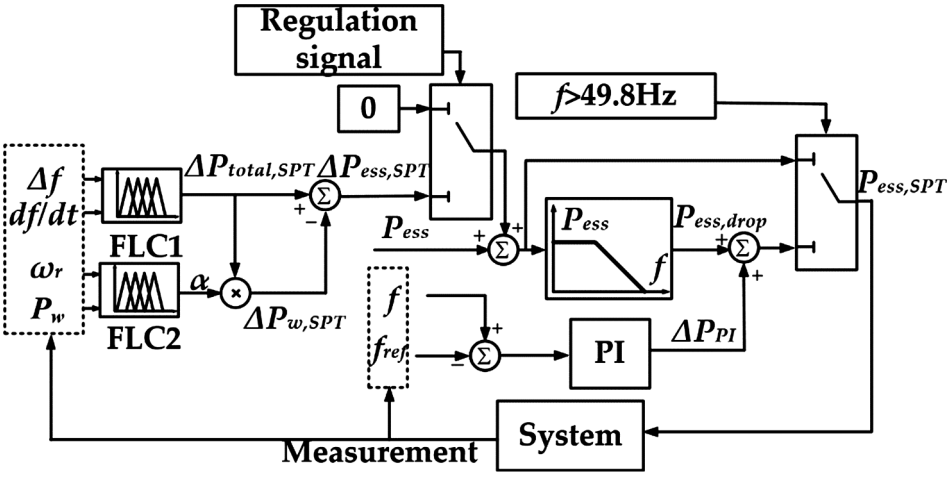


Fig. 6. ES control module diagram.

Based on the realization analysis of DFIG-ES system, the overall control process of various scenarios is as follows:

(1) When the deviation of the system frequency exceeds the allowable range, the regulation flag will be activated and enter the wind speed zone judgement block; otherwise, the DFIG will operate normally in the MPPT mode.
(2) When the current wind speed belongs to the 1st zone, the wind turbine cannot respond to the system frequency fluctuation by reducing its speed and releasing its kinetic energy. If the SOC of the ES is greater than the lower limit, then the DFIG-ES system will respond to the frequency fluctuation only through the ES. Otherwise, it will enter the SOC recovery block.
(3) When the current wind speed belongs to the 2nd zone and the rotor speed of DFIG is larger than $\omega_{r,\min}$, then the active power $P_{w,SPT}$ of the DFIG for frequency response can be determined by FLC2. The main function of ES at this stage is to compensate for the frequency regulation demand that cannot be fulfilled by the active output of the DFIG. When the wind turbine rotor speed decreases and is less than $\omega_{r,\min}$ or the frequency increasing rate is higher than the threshold, then the wind turbine will stop the short-term frequency response process and enter the speed-recovery stage.
(4) When the current wind speed is in the 3rd zone, the rotor speed of the wind turbine will reach its maximum value. Therefore, the rotor kinetic energy of the DFIG in this zone is more sufficient than in the 2nd zone. Although the operation control strategy of this section is the same as that in (3), the power output value of the DFIG is higher than that in the 2nd zone, and the power increment for the frequency fluctuation will be controlled within the upper limit of the wind turbine output.
(5) When the current wind speed belongs to the 4th zone, the output power of the DFIG will reach the rated value. If the rotor kept releasing the kinetic energy at this stage, it would reach the power limit of the wind turbine. Therefore, the wind turbine will maintain the MPPT mode in this zone to make full use of the wind power, and the ES will respond to frequency fluctuations alone.

Furthermore, the application steps of the coordinated control strategy are shown in Fig. 7. In the control cycle, when the real-time frequency exceeds the threshold, the regulation flag will be activated, i.e., step (1), then the DFIG-ES turns into the short-term frequency support mode, i.e., step (2), otherwise the system turns to step (6). The main application steps of the presented control strategy are as follows:

Step (1) When the control cycle begins, the system frequency detection block calculates the real-time Δf and judges whether to activate the regulation flag, if the flag is activated, then it turns to (2), otherwise it turns to (6);
Step (2) FLC1 determines $P_{total,SPT}$, and if the current wind speed belongs to the 2nd/3rd zone, then the combined system turns to (3), otherwise it means the current wind speed belongs to the 1st/4th zone, and it turns to step (6);
Step (3) If the rotor speed $\omega_r > \omega_{min}$, then FLC2 determines α, and $P_{w,SPT}$ can be calculated, afterwards, the combined system turns to (4), otherwise it turns to step 5);
Step (4) In the meantime of calculating $P_{w,SPT}$, if the *SOC* of the ES is larger than its minimum value SOC_{min}, then the ES will cooperate with the DFIG to compensate for the unfulfilled regulation demand, afterwards the combined DFIG-ES system turns to the frequency detection process, otherwise it turn into step (7);
Step (5) The DFIG keeps operating in the MPPT mode, then the system turns to *SOC* judgement in step (4);
Step (6) The DFIG turns into the MPPT mode and if the rotor speed is beyond the normal range, then the system turns into the rotor speed recovery process, otherwise the DFIG-ES system turns to step (8);
Step (7) In the *SOC* recovery block, the ES is controlled to restore its *SOC* to the normal value, which is set as 50%, if the *SOC* recovery process is finished, then it turns to step (8).
Step (8) If the control period is not over, then it turns to back to the frequency detection block in step (6), otherwise this control cycle is finished.

3. Simulation results

3.1. Introduction to the simulation system

To validate the effectiveness of the proposed control strategy, a simulation system is built on the MATLAB/Simulink platform, as shown in Fig. 8. The simulation system is composed of an equivalent DFIG with a capacity of 15 MW, a 2.5 MW/1 MWh ES system, three synchronous generators, transformers and equivalent lines. The wind farm is connected to the 35 kV bus after the 0.69 kV/35 kV transformer. The active load is 20 MW, and a sudden 5 MW load increase will be set at 5 s. The installed capacity of synchronous generators G1, G2 and G3 are 20 MW, 15 MW and 10 MW, respectively.

The synchronous generators in the test system are thermal generators, and the generator model can be seen in Fig. 9.

where ΔP_m is the mechanical power change of the generator, ΔP_L is the load change, ΔP_{tie} is the tie-line power change, R is the droop constant, B is the system frequency response constant, and *ACE* is the ACE signal of the generator. T_g is the time constant of the governor. T_t, T_r and K_r are the time constant of the reheat steam turbine. H is the

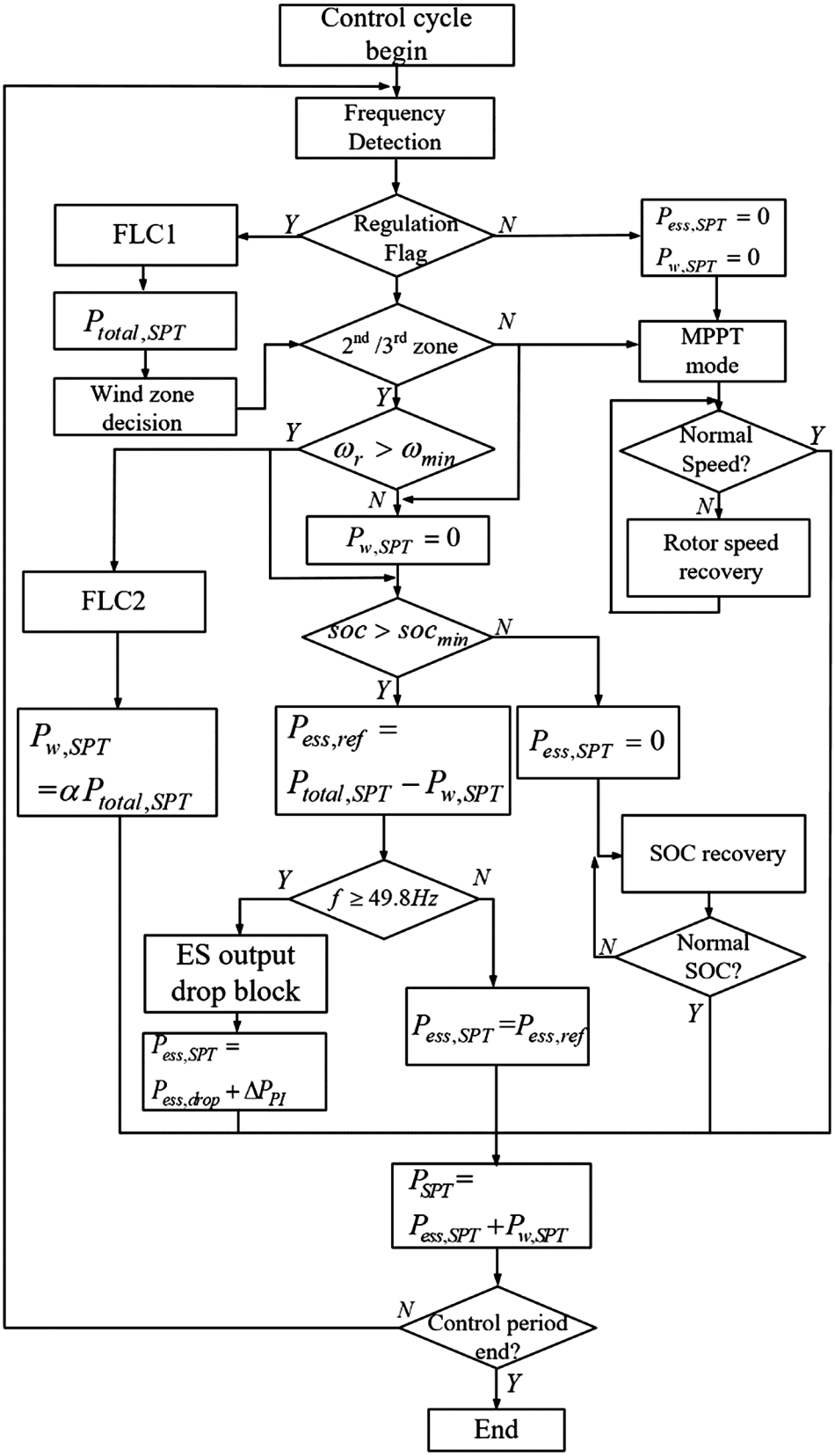


Fig. 7. Application steps of the coordinated control strategy.

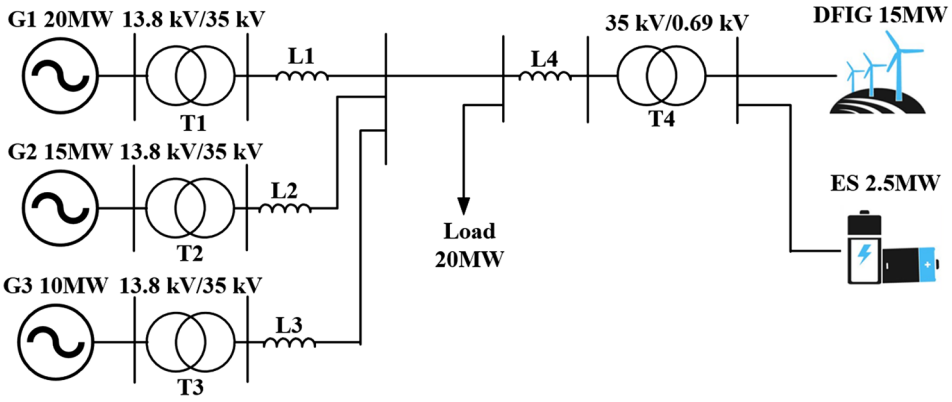


Fig. 8. Diagram of the test power system with a single DFIG-ES system.

generator inertia constant, and D is the load-damping constant. In this paper, $R = 0.05$, $T_g = 0.2$ s, $T_t = 0.3$ s, $T_r = 4$ s, $K_r = 0.3$, $H = 5$ s, $D = 1$.

3.2. Simulation results under different wind speed zones

In this section, to validate the effectiveness of the proposed fuzzy-based control strategy, simulation results of the inertial-loop-based control strategy [5-7], in which only the DFIG participates in frequency support, and the basic cases without any control strategy are chosen for comparison, and their performances under different wind speed zones are analysed.

It is worth noting that when the system operates in the 1st zone, because the rotor speed is unable to drop below its lower limit, only the ES is utilized for frequency support, in a similar manner to the scenario in the 4th zone. Thus, this paper only shows the simulation tests in the 4th zone as the example.

3.2.1. Case study under the 2nd wind speed zone

The simulation wind speed is set as 9 m/s. According to Section 2.1, this wind speed belongs to the 2nd zone. In this wind speed zone, the DFIG output is low, so there is a sufficient power margin for releasing additional power for short-term frequency support. However, due to the low rotor speed in this zone, the rotational kinetic energy stored in the DFIG rotor may be not sufficient to fulfil the frequency regulation requirement. To prevent the rotor speed from dropping below its lower limit, the ES unit is utilized as the backup for the DFIG to compensate for the frequency regulation demand that could not be fulfilled by the rotor kinetic energy of the DFIG.

Fig. 10 shows the simulation results of different control strategies in this scenario, and the data comparison can be seen Table 3. In Fig. 10(a) shows the system frequency response curves. The frequency nadir is 49.63 Hz under the coordinated control strategy, which is higher than the other two control methods (49.55 Hz and 49.46 Hz, respectively). This result shows that the proposed control strategy can effectively increase the nadir of the frequency drop. Under the control method in which only the DFIG participates, a rapid decrease in the output power can be clearly observed when the DFIG exits the regulation process at 15 s. This decrease leads to the second drop in the system frequency, which can also be seen in Fig. 10(a) and is detrimental to the system frequency stability. However, under the proposed control strategy, using the backup of the ES power to compensate for the power shortage caused by the exit of the DFIG, the effect of the second frequency drop has been mitigated, as shown in Fig. 10(a). The curves shown in Fig. 10(b) and (c) are active power output and the rotor speed of the DFIG, respectively. Compared with the inertial-loop-based control strategy, in which only the DFIG participates in the frequency support, the power from the DFIG for frequency support under the proposed control strategy is jointly determined by the participation coefficient and total support power, which are both determined by the FLCs. As can be observed in Fig. 10(b), when the DFIG-ES system operates in the 2nd zone, the participation coefficient is affected by the rotor speed. With the release of kinetic energy, the rotor speed decreases and the coefficient decreases along with it. The frequency regulation demand for the DFIG then also decreases gradually until the end of this process. As a result, the DFIG of the DFIG-only mode undertakes a greater power demand for frequency support, and more rotor kinetic energy have been released for it; thus, the rotor speed declines more than under the other methods. As for the basic mode with no control strategy, the power output of the DFIG keeps operating in the MPPT mode and cannot respond to the system frequency fluctuations. The ES power output is shown in Fig. 10(d). The ES gradually increases its output to compensate for the unfulfilled regulation demand of the DFIG before 15 s. When the DFIG stops supporting the frequency fluctuation at 15 s, the ES quickly increases its output power to suppress the second frequency drop of the system caused by the exit of the DFIG. The ES then gradually decreases its output to avoid excessive power injection along with the frequency gradually returning to the normal range.

The comparison in Fig. 10 shows that the fuzzy-based coordinated control strategy can better satisfy the short-term frequency support requirement than the other two strategies in this zone. After the frequency fluctuation, the frequency drop rate can be effectively restrained, and the frequency nadir of the drop incident increases to a higher value than the other methods.

3.2.2. Case study under the 3rd wind speed zone

In the 3rd wind speed zone, the testing wind speed is set as 11.5 m/s. In this zone, the rotor speed has reached its maximum value, and the kinetic energy stored in the DFIG rotor is more sufficient for short-term frequency support than that of the 2nd zone. However, the power

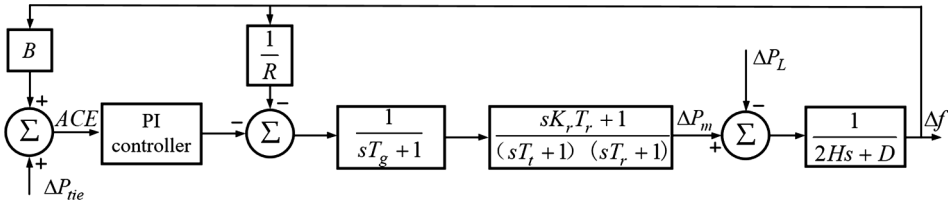


Fig. 9. Model of the thermal generators used in the test power system.

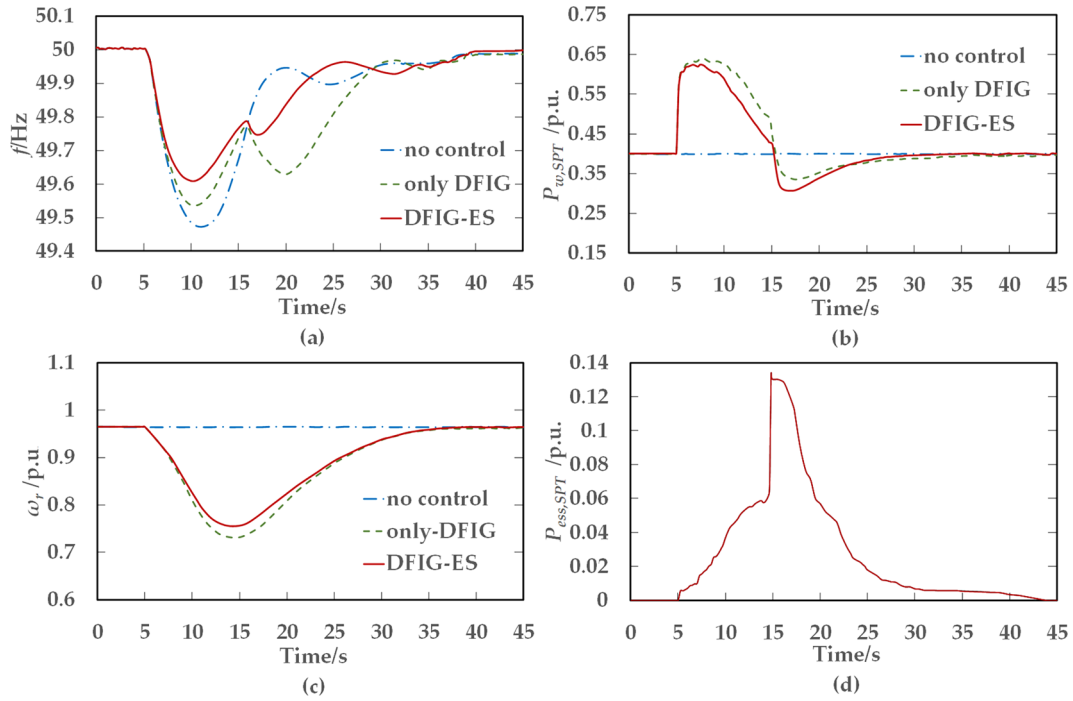


Fig. 10. Simulation results of the 2nd wind zone. (a) Frequency curves of three kinds of control methods, (b) active power output curves of the DFIG, (c) rotor speed curves of the DFIG, and (d) active power output curve of the ES.

Table 3
Simulation results of the 2nd wind zone.

Methods	*f* nadir (Hz)	t_{nadir} (s)	SFD nadir (Hz)
No control	49.46	10.84	
Only DFIG	49.55	10.27	49.61
DFIG-ES	49.63	10.27	49.76

Table 4
Simulation results of the 3rd wind zone.

Methods	*f* nadir (Hz)	t_{nadir} (s)	SFD nadir (Hz)
No control	49.45	10.83	
Only DFIG	49.51	10.26	49.59
DFIG-ES	49.60	10.27	49.72

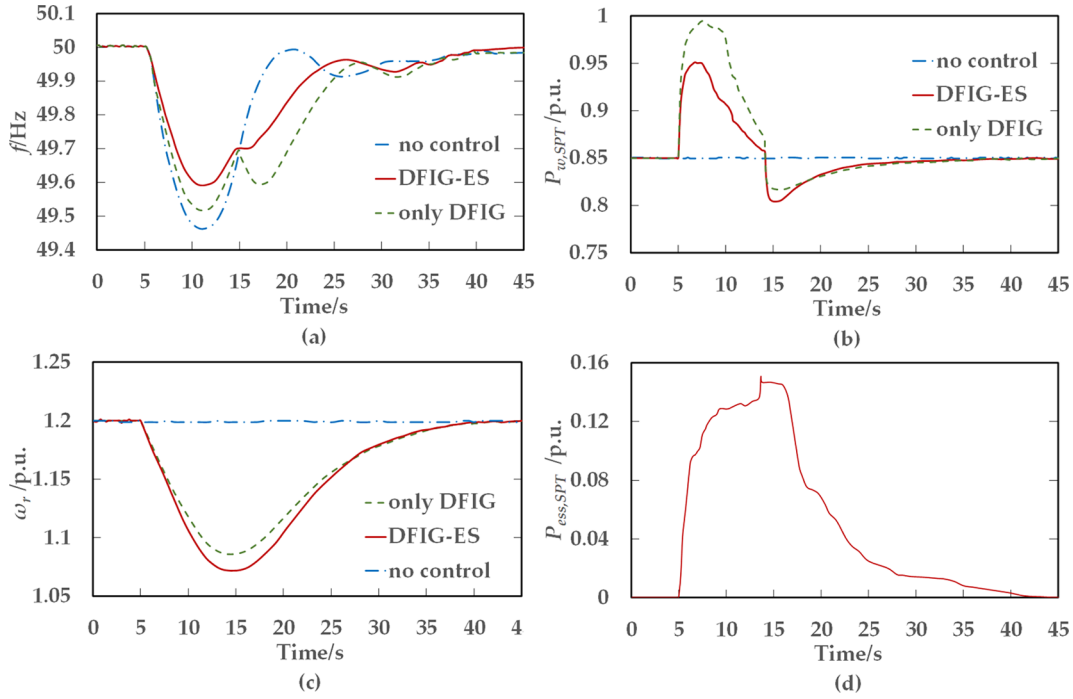


Fig. 11. Simulation results of the 3rd wind zone. (a) Frequency curves of three kinds of control methods, (b) active power output curves of the DFIG, (c) rotor speed curves of the DFIG, and (d) active power output curve of the ES.

output of the DFIG is close to its upper limit, and dynamic cooperation between the DFIG and ES is required to prevent the DFIG power from exceeding the rated limitation.

Fig. 11(a) shows the system frequency curves after the simulation. The simulation results are similar to that of the 2nd zone, and the data comparison can be seen Table 4. The frequency nadir under the proposed control strategy is still higher than that of the DFIG-only method and the method without any control. Moreover, the rate of change in the frequency is also smaller than those of the other methods. The effect of the second frequency drop is also mitigated via the backup of the ES power to compensate for the power shortage caused by the exit of the DFIG, as shown in Fig. 11(a). However, in the DFIG-only method, a rapid decrease in the output power occurs when the DFIG exits the regulation process at 14.9 s. This decrease leads to the second drop of the system frequency, which has negative effects on the frequency stability. Fig. 11(b) and (c) shows the active power and rotor speed curve of the DFIG, respectively. The initial rotor speed is 1.2p.u., which is its maximum value. However, because the DFIG output in this zone is close to its upper limit, to prevent the power from exceeding the limitation, as shown in Fig. 11(b), the DFIG participation coefficient for frequency support gradually decreases to ensure that the output of the DFIG is within the limit. Thus, when the output power of the DFIG is close to the limit before it responds to the frequency fluctuation, the increment in the DFIG output power for frequency support may still have been low even though the kinetic energy is sufficient. Thus, as shown in Fig. 11(d), the ES outputs more energy to compensate for the unfulfilled regulation demand than in Fig. 10(d) when the system frequency begins to drop. Moreover, as shown in Fig. 11(d), the ES also increases its power output after the DFIG quits the regulation process to mitigate the negative effect of the second frequency drop caused by the sudden power imbalance. In the earlier stage of the curve, the ES power is higher than that in Fig. 10(d).

The simulation result validates the ability of ES to effectively cooperate with the DFIG by dynamically adjusting its power output to fulfil the frequency regulation demand. By applying the proposed strategy, the upper limit of the DFIG output is not exceeded, the second frequency drop incident is suppressed and the frequency regulation requirement of the system is fulfilled by the effective cooperation of the DFIG and ES.

3.2.3. Case study under the 4th wind speed zone

In the 4th wind speed zone, the simulation wind speed is set as 12.5 m/s. In this zone, the output power of the DFIG has reached its maximum value, and there is no power margin for abstracting the rotor kinetic energy for frequency support. Due to possibility of exceeding the DFIG power limit, only the ES is applied to respond to the frequency regulation requirement.

Fig. 12 shows the system frequency after the sudden load increase under different control modes, and the data comparison can be seen Table 5. Since the output power of the DFIG-only control mode has reached its upper limit and cannot respond to the frequency fluctuation, the proposed control strategy is compared with only the method without any control. As shown in Fig. 12(a), only the ES responds to the regulation demand in the combined system. After the power injection, the frequency nadir is clearly raised more than that in the non-control mode. Due to the inability of the DFIG kinetic energy to be utilized, the DFIG has no short-term frequency support capability in this zone. As a result, in the power system with a large DFIG integration rate, the system frequency stability in the 4th wind speed zone is weaker than that in the other zones. When frequency incidents occur, the fluctuation amplitude of the system is greater than those of the 2nd and 3rd wind speed zones, and the frequency nadir is lower than those of the other wind zones. Fig. 12(b) shows the ES output curve. When the system frequency returns back above 49.8 Hz, to avoid the detrimental effects caused by excessive ES power injection on the frequency stability, the droop module of the ES is implemented to ensure that the output power of the ES can decrease steadily. After the system frequency exceeds 49.9 Hz, the PI controller is put into use and the additional correction value is added to the output power reference of the ES to keep the system frequency stable near the normal rated value.

Table 5
Simulation results of the 4th wind zone.

Methods	f nadir (Hz)	t_{nadir} (s)
No control	49.43	10.84
DFIG-ES	49.56	10.25

This case proves that the proposed coordinated control strategy can meet the demand of system frequency regulation when the DFIG-ES system operates in the 4th wind speed zone, which cannot be fulfilled by the DFIG-only method. Under the proposed strategy, the system frequency fluctuation is effectively restrained, and the deficiency of the DFIG having no frequency support capability is solved by the dynamic cooperation between the DFIG and ES.

3.2.4. Case study in a varying wind speed condition

In practical operation conditions, the WT always operates with a varying wind speed, instead of always staying in the same wind speed zone. In order to verify the effectiveness of the proposed control strategy in a varying wind speed condition, the test wind speed chosen for this scenario is varying across several wind speed zones. The test wind speed is shown in Fig. 13.

The simulation results under varying wind conditions are shown in Fig. 14, and the data comparison can be seen Table 6. Fig. 14(a) shows the simulation results for the system frequency. Because of the varying power output of the DFIG along with the wind speed, the frequency decreases at the beginning of the simulation. A sudden load increase occurs at 5 s, which aggravates the system frequency dropping rate, causing the frequency to reach the nadir of 49.32 Hz at 12 s. This dropping rate is more serious than under the fixed wind conditions

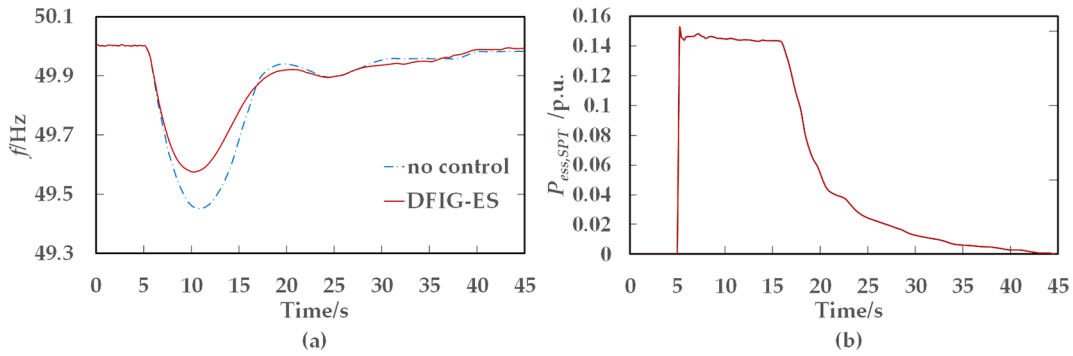


Fig. 12. Simulation results of the 4th wind zone. (a) Frequency curves of two kinds of control methods and (b) active power output curve of the ES.

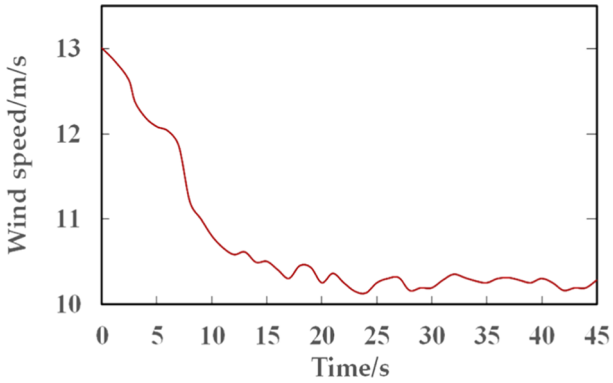


Fig. 13. Wind speed curves of varying wind speeds. Test wind speed for single DFIG-ES system in Section 3.2.4.

considered above. Furthermore, under the proposed control strategy, the frequency nadir is raised from 49.32 Hz to 49.49 Hz, and the frequency drop rate at the beginning stage of the decline in system frequency slowed down. Although the system frequency is still fluctuating with the wind speed, it still gradually recovered to the normal range. Fig. 14(b) and (d) shows the DFIG and ES power output, respectively. When the wind speed entered the 4th zone, as analysed in Section 2, the DFIG does not participate in the frequency regulation; only the ES is put into use when the frequency fluctuation occurs. As the wind speed decreases gradually from the 4th zone to the 3rd zone, the DFIG output margin gradually became sufficient, and the DFIG participation coefficient also increases. Although the DFIG power output decreases with the wind speed, it still responds to the system frequency fluctuation with a relative high participation coefficient and alleviates the burden on the ES in the frequency regulation process. When the DFIG exits the frequency support process, additional power is released from the ES to alleviate the negative influence on system frequency caused by the power imbalance and to help with the rotor speed recovery. Fig. 14(c) shows the rotor speed curve. When the DFIG is out of the frequency regulation, the rotor speed follows the optimal speed reference value at the current wind speed, and thus, the speed is gradually restored.

Table 6
Simulation results in a varying wind speed.

Methods	f nadir (Hz)	t_{nadir} (s)	t_{stable} (s)
No control	49.32	12.1	28.8
DFIG-ES	49.49	11.6	39.1

This case shows that under the application of the proposed control strategy, the frequency regulation demand can be fulfilled by the DFIG-ES system under varying wind speeds, and the short-term frequency support capability of the combined system is adaptable for multi-wind speed situations.

3.2.5. Simulation under different DFIG-ES control strategies

In order to validate the effectiveness and advantage of the proposed FLC-based control strategy for the combined DFIG-ES system, several regular DFIG-ES control methods have been chosen for comparison and the simulation results are shown in Fig. 15 and Table 7.

In method No.1, only DFIG is used, and it is chosen as a reference to show the improvement of other control methods in dealing with the second frequency drop problem. In method No.2, the DFIG operates in 10% de-loading mode and the ES is equipped for auxiliary use. In method No.3, a linear controller is applied in the DFIG-ES system for frequency support and method No.4 represents the proposed control strategy. The simulation conditions are same as in Section 3.2.2.

As shown in Fig. 15(a) and Table 7, all of the DFIG-ES control strategies have a better performance in lifting the frequency nadir than the only-DFIG method. However, as for method No.3, although the second frequency drop nadir is higher than method No.1, it is still much lower than the No.4 method, and the reason is that the DFIG and ES in No.3 method lack the dynamic cooperation. As can be observed in Fig. 15(b), when the DFIG exits the frequency support process and causes the sudden power imbalance, the ES is still following the original output reference instead of increasing its output power to make up for the power imbalance as the No.4 method. As for method No.2, the DFIG rotor kinetic energy is not used for frequency support, so there is no

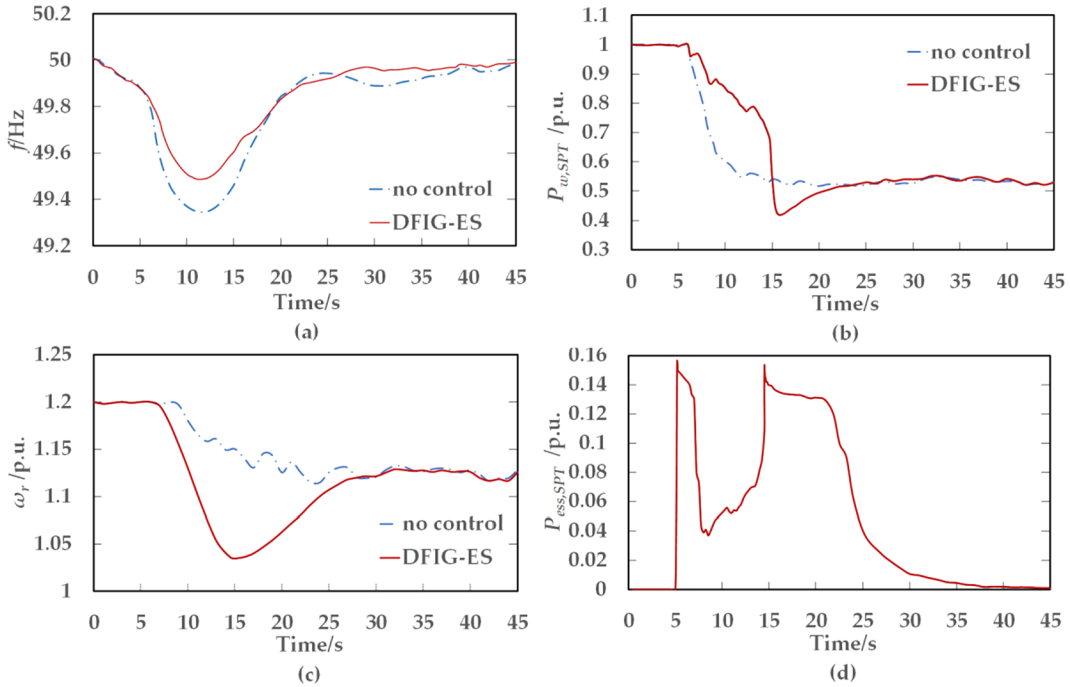


Fig. 14. Simulation results in a varying wind speed (a) Frequency curves of the proposed control method and the basic case, (b) active power output curves of the DFIG, (c) rotor speed curves of the DFIG, and (d) active power output curve of the ES.

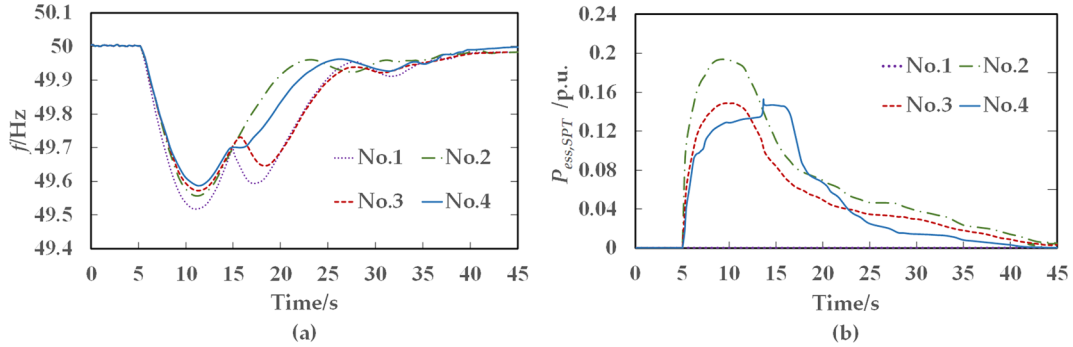


Fig. 15. Simulation results under different control methods for DFIG-ES system (a) Frequency comparison of different control methods, and (b) active power output of the ESs.

Table 7
Simulation results of different DFIG-ES control strategies.

Methods	*f* nadir (Hz)	SFD nadir (Hz)	$P_{ES,p}$ (pu)
No.1	49.51	49.59	\
No.2	49.55	\	0.195
No.3	49.58	49.64	0.149
No.4	49.60	49.71	0.150

second frequency drop problem under this method. However, as shown in Table 7, it has the largest ES output peak value $P_{ES,p}$ among all the methods in the frequency support process, so the capacity of the ES in this method is larger than the others accordingly. In conclusion, the frequency nadir values of the methods from No.2 to No.4 have shown little differences, but from the aspects of second frequency drop problem and the ES capacity requirement, the proposed FLC-based control strategy has better performance than other DFIG-ES control strategies.

3.3. Simulation results of a multi-DFIG-ES system under varying wind conditions

To verify the effectiveness of the proposed control strategy on a power system with multiple wind farms under different wind conditions, a simulation system with three 15 MW-equivalent DFIGs under different varying wind speeds has been built for testing, as shown in Fig. 16. Aside from the DFIGs, the simulation system is composed of two synchronous generators, G1 and G2, whose installed capacities are 80 MW and 60 MW, respectively. Each DFIG is equipped with a 2.5 MW/1 MWh ES unit, and the load is 100 MW. A 15 MW load increase occurs at 5 s.

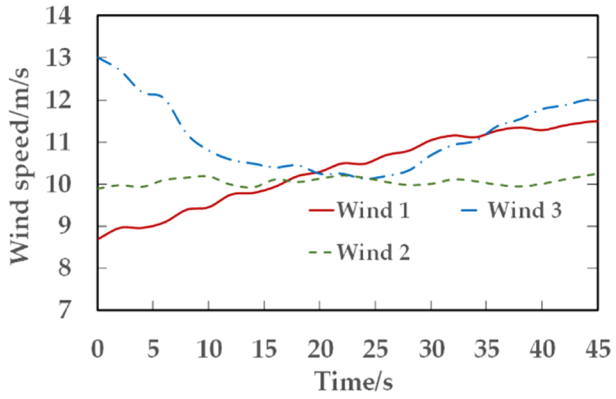


Fig. 17. Wind speed curves of varying wind speeds. Test wind speed for multiple DFIG-ES systems in Section 3.3.

As shown in Fig. 17, Wind 1 for DFIG1 has an increasing trend throughout the simulation process. Wind 2 for DFIG2 shows a gentler trend that fluctuates slightly around 10 m/s. Wind 3 for DFIG3 initially goes down from the 4th zone to the 2nd zone and then increases to the 4th zone again.

Fig. 18(a) shows the simulation results of the overall test system frequency. Due to the varying wind speed, the power outputs of the DFIGs are fluctuating and the frequency slightly decreases at the beginning of the simulation. In Fig. 18(a), the frequency begins to drop at 5 s because of the power imbalance caused by the sudden load increase, and the frequency nadir with no control method is 49.35 Hz at 11 s. However, under the proposed strategy, the frequency nadir is lifted to

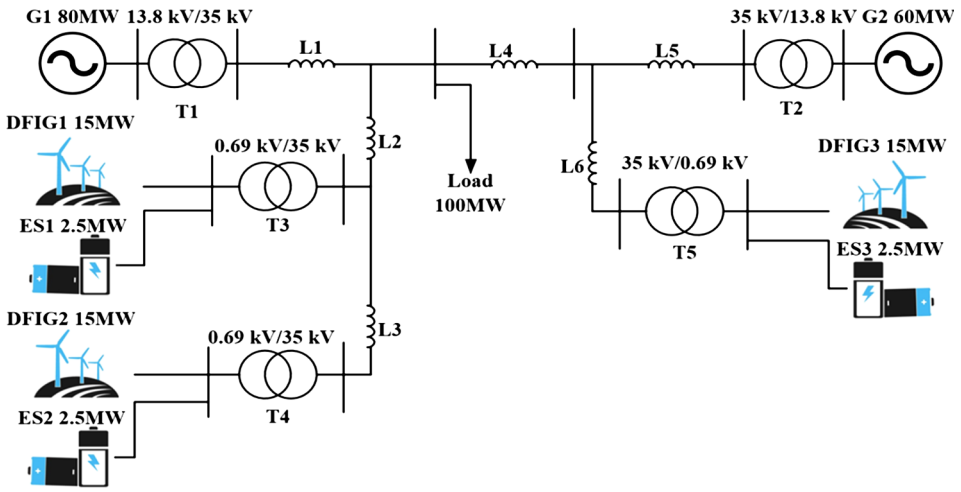


Fig. 16. Diagram of the test power system with multiple DFIG-ES systems.

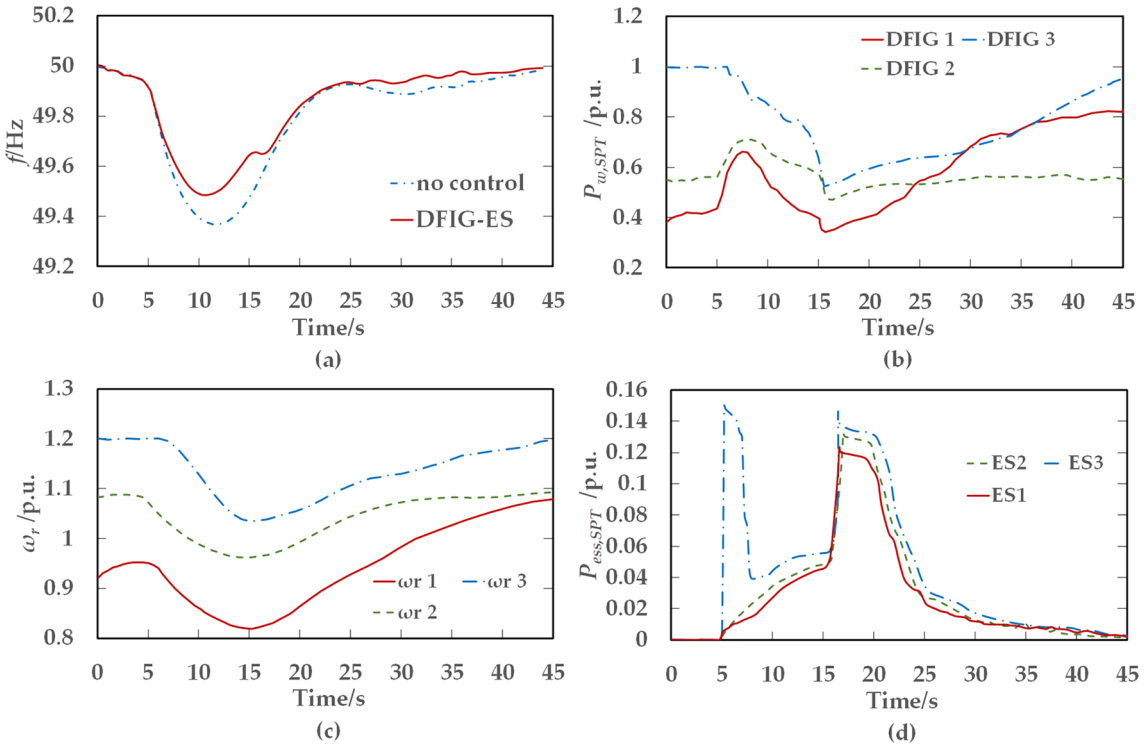


Fig. 18. Simulation results of multiple DFIG-ES systems under varying wind conditions (a) Frequency curves comparison of two kinds of control methods, (b) active power output curves of the DFIGs, (c) rotor speed curves of the DFIGs, and (d) active power output curve of the ESs.

49.51 Hz, and the frequency change rate is smaller than with the other methods. Fig. 18(b) shows the power output curves of the three DFIGs. The DFIGs responds to the frequency fluctuation according to the participation coefficient determined by the FLCs. The difference between this simulation and the previous simulation is that after the frequency response process, the DFIG output follows the varying reference value along with the varying wind speed instead of maintaining a constant value. Fig. 18(c) shows the rotor speeds of the three DFIGs. For rotor 1, before 5 s, its speed follows its MPPT reference with increasing wind speed until it has to decrease to release its kinetic energy. When the DFIGs stops responding to the frequency fluctuation at 15.2 s, the rotor speed begins to return to the MPPT mode and then follows its MPPT reference value again. Meanwhile, the rotor speed keeps increasing along with the increasing wind speed. The same trend occurs for the other two curves, although the rotor speed of third DFIG remains at 1.2p.u. initially because it stays within the 3rd and 4th zones before 5 s. The curves of the power outputs of the ESs are shown in Fig. 18(d). The power values increase at 15.2 s to mitigate the second frequency drop made by the exit of the DFIG from the frequency support. ES output 3 shows a higher value between 5 s and 8 s because the wind speed during this time interval belongs to the 4th zone, and the DFIG still cannot increase its output for frequency support because of its upper power limit. Then, after 8 s, the wind speed drops into the 3rd zone, and ES output 3 becomes smaller because the DFIG can once again respond to the frequency fluctuations.

This paper aims to present a novel control and operation strategy of the DFIG and ES unit under various wind speeds and verify its effectiveness. In a power grid with a high wind power penetration rate, the WTs are widely distributed, and due to the different geographical and wind speed conditions, the operation states of the DFIGs are always different from each other, even on the same wind farm. Through the simulation of the multi-DFIG scenario in Fig. 15 and based on the fuzzy-based control strategy and wind speed zone classification proposed in this paper, the short-term frequency support capability of each DFIG in the multi-DFIG integrated power grid has been effectively improved.

4. Conclusion

In this paper, a novel fuzzy-logic based coordinated control strategy is presented for improving the short-term frequency support capability of the DFIG-ES system. To demonstrate the effectiveness of the proposed strategy, test systems including the combined DFIG-ES system and synchronous generators are built in MATLAB/Simulink platform, and the performance of the frequency support is evaluated by case studies. Compared with the conventional only-DFIG control methodologies under various wind speed scenarios and multiple DFIG-ES system scenario, the better performance of the proposed control strategy for short-term frequency support are demonstrated. Under the proposed strategy, the DFIG and ES could cooperate dynamically for short-term frequency support and improve the system frequency stability without de-loading operation, and the secondary frequency drop problem in the frequency restoration process can be alleviated, moreover, with considering the wind speed zone classification in the design process, the DFIG-ES system can provide frequency support in a wider range of wind speed conditions with better reliability and adaptability.

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Appendix A

The main parameters of the test power systems used in this paper are following parameters are obtained from [22,24] (see Tables A1–A4).

Table A1
Parameters of the system line in Fig. 9.

Line	R (Ω/km)	X (Ω/km)	B (S/km)	Length (km)
L1	0.065	0.315	3.65×10^{-6}	10
L2	0.065	0.315	3.65×10^{-6}	10
L3	0.065	0.315	3.65×10^{-6}	10
L4	0.065	0.315	3.65×10^{-6}	15

Table A2
Parameters of the system line in Fig. 15.

Line	R (Ω/km)	X (Ω/km)	B (S/km)	Length (km)
L1	0.065	0.315	3.65×10^{-6}	10
L2	0.065	0.315	3.65×10^{-6}	15
L3	0.065	0.315	3.65×10^{-6}	10
L4	0.065	0.315	3.65×10^{-6}	15
L5	0.065	0.315	3.65×10^{-6}	10

Table A3
Parameters of DFIG.

R_s	R_r	X_{ls}	X_{lr}	X_{lm}	H
0.058 p.u.	0.105 p.u.	0.087 p.u.	0.118 p.u.	3.78 p.u.	5.75 s

Table A4
Parameters of the synchronous generator.

x_d	x'_d	x''_d	x_q	x'_q	x''_q	T'_{d0}
1.8 p.u.	0.1 p.u.	0.25 p.u.	1.7 p.u.	0.55 p.u.	0.25 p.u.	8 s
T''_{d0}	T'_{q0}	T''_{q0}	R_a	K_D	H	
0.03 s	0.4 s	0.05 s	0.0025 p.u.	1	5 s	

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